CIS41032: Practical for Advance Database Management System

Department of ICT Faculty of Technology
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Lab Sheet 04

Title: Update, Delete Commands in MongoDB.

Objective:

- Understand about Update operations.
- Understand delete commands.

Dataset: Student Records

StudentID	Name	Age	Grade	Section	Marks (%)	EnrollmentDate	Status
101	Emma Johnson	14	9	A	85	2021-06-10	Active
102	Liam Smith	15	10	B	90	2020-09-12	Active
103	Olivia Brown	13	8	A	88	2022-02-15	Active
104	Noah Wilson	16	11	C	75	2019-07-20	Graduated
105	Ava Miller	14	9	B	92	2021-01-18	Active
106	James White	17	12	C	80	2018-05-25	Graduated
107	Charlotte Lee	15	10	A	87	2020-08-30	Active
108	Ethan Clark	16	11	B	78	2019-09-10	Active
109	Mia Walker	13	8	C	89	2022-03-11	Active
110	Henry Adams	17	12	A	82	2018-04-05	Graduated

- Creates database **SchoolDB**
- Creates the collection as **"Students"**
- Insert all data.

<u>Update Operations</u>

1. Increase the marks of all students in **Grade 10** by 5%.
2. Change the **status** of students in **Grade 12** to "**Alumni**".
3. Rename the "**Marks (%)**" field to "**Score**" in all records.
4. Update the **section** of "**Emma Johnson**" from "**A**" to "**B**".
5. Add a new field "**Scholarship**" for students who scored **above 90%**, with a value of "**Yes**".

Delete Operations

1. Delete students who have the status
2. Remove students from **Grade 11** who enrolled **before 2020**.
3. Delete a specific student record for "**James White**".
4. Remove the "**Scholarship**" field from all students.

Exercise 01

Dataset: Books Collection

BookID	Title	Author	Genre	Published Year	Copies Available	Total Copies	Status
301	The Great Gatsby	F. Scott Fitzgerald	Fiction	1925	3	5	Available
302	To Kill a Mockingbird	Harper Lee	Fiction	1960	1	4	Available
303	1984	George Orwell	Dystopian	1949	0	6	Borrowed
304	The Catcher in the Rye	J.D. Salinger	Fiction	1951	2	3	Available
305	Moby-Dick	Herman Melville	Adventure	1851	5	5	Available
306	Pride and Prejudice	Jane Austen	Romance	1813	1	4	Available
307	The Hobbit	J.R.R. Tolkien	Fantasy	1937	0	7	Borrowed
308	War and Peace	Leo Tolstoy	Historical	1869	2	5	Available
309	Hamlet	William Shakespeare	Drama	1603	4	6	Available
310	Crime and Punishment	Fyodor Dostoevsky	Crime	1866	3	4	Available

1. Create a new database called **LibraryDB.**
2. Create a collection named **Books**.
3. Insert all the book records from the dataset above.
4. Increase the total copies of "**1984**" by **2** and adjust available copies accordingly.
5. Mark "**Pride and Prejudice**" as "**Reserved" if copies available are less than 2.**
6. Update all books published **before 1900** to have a **new field "Classic**": true.

7. Change the **genre of "The Hobbit**" to "**Epic Fantasy**".
8. Add a "**LastBorrowedDate**" field to books that are currently borrowed (set a sample date).
9. Delete all books where **TotalCopies** is **less than 3** (outdated books removal).
10. Remove the "**Crime and Punishment**" book from the database.
11. Remove the "**Classic"** field from books **published after 1900** (to correct the dataset).
12. Delete books that have been **borrowed but have 0 total copies left**.